

Cole Zill

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SUMMARY

Software Engineer with expertise in secure systems and infrastructure security, particularly in aerospace and AI-enabled environments. Proven ability to design and implement security mechanisms for production systems, leveraging skills in Python, CI/CD, and system hardening. Experienced in embedding security into distributed and safety-critical systems, developing automation tools, and enhancing system resilience. Passionate about distributed systems, cloud infrastructure, and large-scale security engineering.

EXPERIENCE

Software/Systems Engineer I

Honeywell Aerospace

January 2026 - Present, Phoenix, AZ

- Developed and maintained software systems using MATLAB and .NET in safety-critical environments, ensuring high reliability and security compliance.
- Applied secure coding practices to legacy and embedded systems, reducing vulnerability exposure and improving system integrity.
- Collaborated with cross-functional teams to integrate security requirements into system architecture and software design.
- Contributed to secure software lifecycle practices while delivering production features under strict regulatory constraints.
- Built internal tooling or automation scripts in Python to support development and validation workflows.
- Contributed to system-level design discussions, integrating security controls into distributed and embedded system architectures.

Cyber Security Architect/Engineer I

Honeywell Aerospace

June 2024 - January 2026, Phoenix, AZ

- Engineered and automated security workflows by integrating ServiceNow and Jira APIs, reducing manual triage and improving incident response efficiency.
- Led end-to-end security engineering across AI/ML-enabled systems, including threat modeling, vulnerability remediation, and secure deployment of third-party libraries.
- Built and enforced secure development pipelines using SAST (Coverity), shifting vulnerability detection earlier in CI/CD and reducing downstream risk.
- Designed and implemented security controls aligned with OWASP and aerospace standards, strengthening system resilience across avionics platforms.
- Partnered with software and systems engineers to embed security mechanisms directly into production systems, enabling secure-by-design development.
- Conducted security assessments and risk modeling on distributed avionics systems, identifying and mitigating critical attack surfaces.

INVOLVEMENT

Alumi Cybersecurity/Software Speaker

Maricopa County Community Colleges • Paradise Valley Community College • May 2022 - May 2024

- Discussed practical and engaging cybersecurity labs to current students in the program.
- Engaged with fellow students on resume building as well as real time projects to showcase skills learned from coursework such as honeypot setups via cloud environments, ethical hacking principles against local vulnerable machines, and encryption techniques to protect data in transit and at rest.
- Gave numerous lectures during STEM conventions on how to land a role in Cybersecurity for current students, and also how to stand out from other junior applicants via projects, internships and contributing to open source software.

SKILLS

Python, C#, C, C++, Java, Go, SQL, Bash, Powershell

Secure Software Development (SSDLC), Threat Modeling (MSTM), PKI, System Hardening, Vulnerability Management

CI/CD, APIs, Automation, Version Control

SAST (Coverity), OWASP, Risk Assessment, SCA (BlackDuck), Continuous Monitoring, Risk Scoring

Embedded Systems, Distributed Systems, Avionics Software, Scalable Systems, AI Agents, LLMs, AI Security

PROJECT

Mi-Agent — Secure Local-First AI Agent Framework

- Built a production-grade AI agent framework in Python (~11k+ LOC) for autonomous task execution on constrained systems (e.g., Raspberry Pi).
- Engineered defense-in-depth security controls for LLM tool execution, including sandboxed Python execution, shell filtering, workspace isolation, and resource limits.
- Developed secure orchestration for file, shell, and web tools with input validation and mitigations against prompt injection, command injection, and privilege escalation.
- Implemented distributed agent capabilities including async event bus, multi-agent orchestration, dynamic model routing, and failover handling.
- Designed authentication, access control, semantic memory, and telemetry systems with SQLite-backed observability and external channel protection.

EDUCATION

B.S. Artificial Intelligence & Machine Learning

Chandler Gilbert College • Phoenix, AZ • December 2026 • 3.9
• – Phi Theta Kappa Scholarship Award Recipient (GPA > 3.5).

A.A.S. Cybersecurity/Cyber Engineering

Paradise Valley Community College • Phoenix, AZ • May 2024
